

While Loop Diagnostic Test Solutions

You must show all your working in the following. No correct table (exactly the same as the solution)... no marks.

1. What is printed when the following code fragment has executed?

```
int t = 3;                // L1
while (t <= 7)            // L2
{
    t = t + 2;            // L3
    System.out.println(t); // L4
}
System.out.println(t);    // L5
```

Printed:

5
7
9
9

LINE	t	t<=7?	Printed
L1	3		
L2		3<=7? true	
L3	5		
L4			5
L2		5<=7? true	
L3	7		
L4			7
L2		7<=7? true	
L3	9		
L4			9
L2		9<=7? false	
L5			9

2. What is printed when the following code fragment has executed?

```
int t = 8;                // L1
while (t <= 7)            // L2
{
    t = t + 2;            // L3
    System.out.println(t); // L4
}
System.out.println(t);    // L5
```

Printed:

8

LINE	t	t<=7?	Printed
L1	8		
L2		8<=7? false	
L5			8

3. What is printed when the following code fragment has executed?

```
int t = 7; // L1
while (t <= 7) // L2
{
    System.out.print(t); // L3
    t = t + 2; // L4
}
System.out.println(t); // L5
```

Printed:

79

LINE	t	t<=7?	Printed
L1	7		
L2		7<=7? true	
L3			7
L4	9		
L2		9<=7? false	
L5			9

4. What is printed when the following code fragment has executed?

```
int x = 2; // L1
while (x < 8) // L2
{
    x = x + 3; // L3
}
System.out.println(x); // L4
```

Printed:

8

LINE	x	x<8?	Printed
L1	2		
L2		2<8? true	
L3	5		
L2		5<8? true	
L3	8		
L2		8<8? false	
L5			8